SCHOOL OF COMPUTATION, INFORMATION AND TECHNOLOGY — INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Bachelor's Thesis in Informatics

Investigation of Convolutional Neural Networks for Iterative Solver Selection

Leon Unterberger

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Investigation of Convolutional Neural Networks for Iterative Solver Selection

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I confirm that this bachelor's thesis is my own work and I have documented all sources and material used.

Munich, 23.07.2026 Leon Unterberger

Acknowledgments

Abstract

Contents

1 Introduction

1.1 Motivation

Linear equations are a construct that one can observe in a lot of different fields. There are used in simple examples already in school and get really important when working with computer vision, robotics, machine learning [GQ19], 3D transformations, projections and also in biology. While the linear equations in school are easily solvable by applying direct approaches, in harder problems the Cost of O(n^3) hits the ceiling of usability pretty fast and iterative solver are used. They have a Cost of

1.2 Section

Citation test [Lam94].

Acronyms must be added in `main.tex` and are referenced using macros. The first occurrence is automatically replaced with the long version of the acronym, while all subsequent usages use the abbreviation.

E.g. `\ac{TUM}, \ac{TUM}` $\Rightarrow$ Technical University of Munich (TUM), TUM

For more details, see the documentation of the `acronym` package[1].

1.2.1 Subsection

See Table 1.1, Figure 1.1, Figure 1.2, Figure 1.3.

Table 1.1: An example for a simple table.

A	B	C	D
1	2	1	2
2	3	2	3

[1] `https://ctan.org/pkg/acronym`

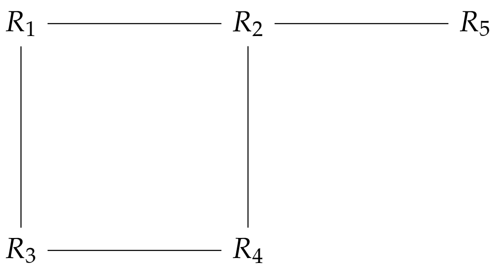

Figure 1.1: An example for a simple drawing.

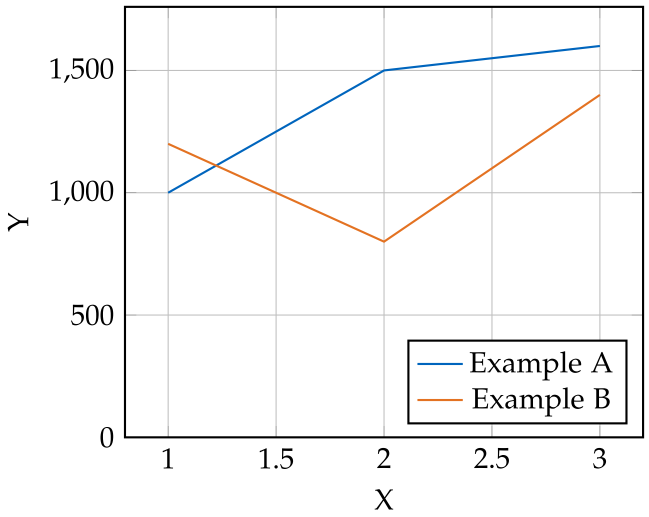

Figure 1.2: An example for a simple plot.

```
SELECT * FROM tbl WHERE tbl.str = "str"
```

Figure 1.3: An example for a source code listing.

Abbreviations

TUM Technical University of Munich

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Bibliography

[GQ19] J. Gallier and J. Quaintance. "Linear algebra for computer vision, robotics, and machine learning." In: *University of Pennsylvania* (2019).

[Lam94] L. Lamport. *LaTeX : A Documentation Preparation System User's Guide and Reference Manual*. Addison-Wesley Professional, 1994.